Choosing Factors

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Introduction

Main Goal

- Our goal is to develop insights about the maximum squared Sharpe ratio for model factors as a metric for ranking asset pricing models.
- $Sh^2(f) = \mu_f' \Sigma_f^{-1} \mu_f$

Nested Model

- Def: One model's factor are a subset of another's model. (e.g. $CAPM \subset ThreeFactor \subset FiveFactors \subset SixFactor$)
- Question: Does adding a factor help?

Non-Nested Model

- Def: Models or factors are independent with each other
- Question: Which set of factors is better?

Nested vs. Non-Nested Factor Models

Nested Models

CAPM	**Factors**
Three-factor model	MKT + SMB + HML
Five-factor model	MKT + SMB + HML + RMW + CMA
Six-factor model	MKT + SMB + HML + RMW + CMA + UMD

Non-Nested Models Three issues when comparing a six-factor model:

1. Cash profitability vs. operating profitability
2. Long–short spread factors vs. excess-return factors
3. Whether to use only small stocks, only big stocks, or both

Previous Approaches

1. LHS Approach — Pricing Errors

Method: Regress test-portfolio excess returns on model factors and evaluate the intercepts α. Smaller α implies better pricing.

Limitation: Model rankings can depend on the choice of test portfolios; changing the LHS assets may change the conclusion.

2. RHS Approach — Spanning Regressions

Method: Test whether a candidate factor is spanned by existing factors:

$$f_{\text{new},t} = a + \sum_k b_k f_{k,t} + e_t.$$

A significant a suggests that the candidate factor adds information.

Limitation: Works naturally for nested models, but cannot directly compare non-nested models with distinct competing factors.

Maximum Squared Sharpe Ratio

Assets: N, weights: w, $[r] = \mu$, $(r) = \Sigma$

For a portfolio w,

$$Sh(w) = \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}}, \qquad Sh^2(w) = \frac{(w^\top \mu)^2}{w^\top \Sigma w}.$$

$$(w^\top \mu)^2 = \left(w^\top \Sigma^{1/2} \Sigma^{-1/2} \mu\right)^2$$

$$\text{(Cauchy–Schwarz inequality)} \leq (w^\top \Sigma w)(\mu^\top \Sigma^{-1} \mu).$$

Therefore,

$$\boxed{\max_w Sh^2(w) = \mu^\top \Sigma^{-1} \mu}$$

with optimal portfolio weights

$$\boxed{w^* \propto \Sigma^{-1} \mu.}$$

Using Maximum Sharpe Ratio to Measure Model i

Suppose model i explains returns Π_i using factors f_i:

$$\Pi_{i,t} = a_i + B_i f_{i,t} + e_{i,t}.$$

Barillas–Shanken (BS; 2017): The winning model minimizes the unexplained maximum squared Sharpe ratio:

$$\min_i Sh^2(a_i) = a_i^\top \Sigma_i^{-1} a_i$$

Gibbons–Ross–Shanken (GRS; 1989):

$$Sh^2(a_i) = Sh^2(\Pi_i, f_i) - Sh^2(f_i).$$

If $\Pi_i \cup f_i = \Pi$ is the same total opportunity set for every model

$$Sh^2(a_i) = Sh^2(\Pi) - Sh^2(f_i).$$

Therefore,

$$\boxed{\min_i Sh^2(a_i) \iff \max_i Sh^2(f_i).}$$

Estimates of $Sh^2(f)$ are upward biased. For nested models, spanning regressions (t-tests / GRS tests) account for sampling error. For non-nested models, raw $\widehat{Sh^2(f)}$ cannot be compared directly; Fama–French therefore use bootstrap simulations.

Bootstrap Simulation

Data Sample: July 1963–June 2016: 636 months $\Rightarrow$ 318 adjacent pairs

$$(1,2),(3,4),\ldots,(635,636).$$

For each simulation run:

- Draw 318 pairs **with replacement**.
- Randomly assign one month from each pair to in-sample (IS) and the other to out-of-sample(OS):

$$(N, N+1) \quad \Rightarrow \quad N \in IS,\ N+1 \in OS$$

- **IS:** estimate optimal factor weights under $\max \mathrm{Sh}^2(f)$.
- **OS:** apply the IS weights to the unused months and calculate $\mathrm{Sh}^2(f)_{\mathrm{OS}}$.
  $Sh^2(f)_{\mathrm{OS}}$ is free of in-sample optimization bias.
- FS: resample all 636 months with replacement.

The Six-Factor Model

$$R_{it} - F_t = a_i + b_i Mkt_t + s_i SMB_t + h_i HML_t + r_i RMW_{O,t} + c_i CMA_t + m_i UMD_t + e_{it}.$$

R_{it}	return on asset i in month t
F_t	one-month Treasury bill rate (risk-free rate)
a_i	regression intercept (pricing error / alpha)
$b_i, s_i, h_i, r_i, c_i, m_i$	factor loadings
e_{it}	regression residual
Mkt_t	value-weighted market excess return
SMB_t	small minus big
HML_t	high minus low book-to-market
$RMW_{O,t}$	robust minus weak operating profitability
CMA_t	conservative minus aggressive investment
UMD_t	up minus down (momentum)

Three Non-Nested Factor-Choice Problems

Problem 1: Operating vs. Cash Profitability

$$RMW_O \quad \text{vs.} \quad RMW_C$$

Operating profitability (OP):	includes accruals
Cash profitability (CP):	excludes the effect of accruals

Problem 2: Small + Big vs. Small Stocks Only

Standard Way: underestimate premium in small stocks and overestimate premium in big stocks

$$HML = \frac{HML_S + HML_B}{2}$$

Alternative: keep Mkt and SMB, but use only small-stock components of HML, CMA, RMW, UMD.

Problem 3: Spread Factors vs. Excess-Return Factors

$$\text{Standard:} \qquad HML = H - L \qquad \text{(long–short spread)}$$

$$\text{Alternative:} \qquad H - F \quad \text{or} \quad L - F,$$

where F is the risk-free rate.

Results for Nested Models

One added factor: use the t-statistic of a_i. Multiple added factors: use the GRS test.

Table 2

Spanning tests for nested models: July 1963–June 2016.

This table tests whether the excess market return, *Mkt*, spans the size spread factor, *SMB*, and the value spread factor, *HML;* whether *Mkt*, *SMB*, and *HML* span the investment spread factor, *CMA*, and a profitability spread factor, RMW_O or RMW_C; and whether *Mkt*, *SMB*, *HML*, *CMA*, and RMW_O or RMW_C span the momentum spread factor, *UMD*. The tests center on the intercepts from spanning regressions of additional factors on base factors: *SMB* and *HML* regressed on *Mkt; CMA* and RMW_O or RMW_C on *Mkt*, *SMB*, and *HML*; and *UMD* on *Mkt*, *SMB*, *HML*, *CMA*, and RMW_O or RMW_C. Panel A shows regression coefficients and *t*-statistics for the coefficients. Panel B shows *GRS* statistics from Gibbons et al. (1989) and their *p*-values testing whether the additional factors jointly improve the maximum squared Sharpe ratio, $Sh^2(f)$, produced by the factors of the base model, either the capital asset pricing model (CAPM) or the three-factor model. The *t*-statistics for the regression intercepts provide tests for individual additional factors.

Panel A: Spanning regressions

	Coefficient							*t*(Coefficient)								
LHS	*Int*	*Mkt*	*SMB*	*HML*	RMW_O	RMW_C	*CMA*	*Int*	*Mkt*	*SMB*	*HML*	RMW_O	RMW_C	*CMA*	R^2	$s(e)$
SMB	0.17	0.18						1.46	6.89						0.07	2.93
HML	0.43	−0.17						4.01	−6.89						0.07	2.71
RMW_O	0.34	−0.07	−0.23	0.01				4.01	−3.70	−8.16	0.25				0.14	2.07
RMW_C	0.48	−0.14	−0.26	0.06				7.88	−10.12	−13.05	2.70				0.39	1.49
CMA	0.20	−0.10	0.01	0.46				3.53	−7.57	0.35	22.34				0.52	1.39
UMD	0.73	−0.13	0.08	−0.54	0.25		0.41	4.34	−3.07	1.37	−6.74	3.12		3.47	0.09	4.05
UMD	0.61	−0.09	0.14	−0.53		0.46	0.34	3.55	−1.97	2.30	−6.72		4.29	2.91	0.10	4.02

Panel B: Multi-factor tests

Model	LHS returns	*GRS*	*p*-value
CAPM	*SMB, HML*	9.20	0.000
Three-factor model	RMW_O, *CMA*	17.99	0.000
Three-factor model	RMW_C, *CMA*	37.46	0.000

Table 2: Spanning regressions and GRS tests for nested models.

Result for Problem 1: RMW_O vs. RMW_C

- Consider two sets of six-factor models: only difference in RMW_O or RMW_C
- FS and IS are upward biased, and $means$ and $medians$ of $Sh^2(f)$ are higher in IS than FS (since sampling error of $Sh^2(f)$ in IS $>$ that in FS)

Panel A: Levels of $Sh^2(f)$

		Full-sample		In-sample		Out-of-sample	
Model	Actual	Average	Median	Average	Median	Average	Median
Six-factor Operating Profitability							
Mkt, SMB, HML, RMW_O, CMA, UMD	0.135	0.152	0.149	0.177	0.169	0.108	0.102
Mkt, SMB, HML_S, RMW_{OS}, CMA_S, UMD_S	0.199	0.217	0.213	0.244	0.236	0.169	0.162
Mkt, S–F, H–F, R_O–F, C–F, U–F	0.134	0.148	0.146	0.172	0.166	0.106	0.100
Mkt, S–F, H_S–F, R_{OS}–F, C_S–F, U_S–F	0.167	0.182	0.180	0.206	0.200	0.138	0.132
Mkt, S–F,L–F, W_O–F, A–F,D–F	0.111	0.127	0.124	0.152	0.144	0.085	0.079
Mkt, S–F, L_S–F, W_{OS}–F, A_S–F, D_S–F	0.174	0.192	0.189	0.220	0.212	0.144	0.137
Six-factor cash profitability							
Mkt, SMB, HML, RMW_C, CMA, UMD	0.190	0.208	0.205	0.236	0.228	0.159	0.152
Mkt, SMB, HML_S, RMW_{CS}, CMA_S, UMD_S	0.226	0.244	0.241	0.274	0.265	0.194	0.186
Mkt, S–F, H–F, R_C–F, C–F, U–F	0.177	0.193	0.191	0.218	0.212	0.147	0.141
Mkt, S–F, H_S–F, R_{CS}–F, C_S–F, U_S–F	0.188	0.204	0.201	0.229	0.221	0.159	0.153
Mkt, S–F,L–F, W_C–F, A–F,D–F	0.160	0.176	0.173	0.204	0.196	0.128	0.121
Mkt, S–F, L_S–F, W_{CS}–F, A_S–F, D_S–F	0.210	0.229	0.226	0.259	0.250	0.177	0.170

Panel B: Differences between $Sh^2(f)$ for cash and operating profitability factors

		Full-sample			In-sample			Out-of-sample		
Model	Actual	Average	Median	% < 0	Average	Median	% < 0	Average	Median	% < 0
Cash – Operating Profitability										
Mkt, SMB, HML, RMW, CMA, UMD	0.055	0.056	0.054	0.0	0.059	0.055	1.1	0.050	0.047	2.2
Mkt, SMB, HML_S, RMW_S, CMA_S, UMD_S	0.027	0.028	0.027	1.0	0.030	0.027	8.2	0.025	0.023	10.0
Mkt, S–F, H–F, R–F, C–F, U–F	0.043	0.045	0.043	0.0	0.047	0.044	1.1	0.041	0.039	2.0
Mkt, S–F, H_S–F, R_S–F, C_S–F, U_S–F	0.021	0.022	0.021	4.5	0.023	0.020	17.3	0.022	0.020	15.4
Mkt, S–F,L–F, W–F, A–F,D–F	0.048	0.049	0.047	0.2	0.052	0.047	4.1	0.043	0.040	5.0
Mkt, S–F, L_S–F, W_S–F, A_S–F, D_S–F	0.036	0.037	0.036	0.0	0.039	0.035	1.8	0.033	0.031	5.4

Results for Problem 2 and 3

Panel A: Levels of $Sh^2(f)$

		Full-sample		In-sample		Out-of-sample	
Model	Actual	Average	Median	Average	Median	Average	Median
Six-factor Operating Profitability							
Mkt, SMB, HML, RMW_O, CMA, UMD	0.135	0.152	0.149	0.177	0.169	0.108	0.102
Mkt, SMB, HML_S, RMW_{OS}, CMA_S, UMD_S	0.199	0.217	0.213	0.244	0.236	0.169	0.162
Mkt, S–F, H–F, R_O–F, C–F, U–F	0.134	0.148	0.146	0.172	0.166	0.106	0.100
Mkt, S–F, H_S–F, R_{OS}–F, C_S–F, U_S–F	0.167	0.182	0.180	0.206	0.200	0.138	0.132
Mkt, S–F,L–F, W_O–F, A–F,D–F	0.111	0.127	0.124	0.152	0.144	0.085	0.079
Mkt, S–F, L_S–F, W_{OS}–F, A_S–F, D_S–F	0.174	0.192	0.189	0.220	0.212	0.144	0.137
Six-factor cash profitability							
Mkt, SMB, HML, RMW_C, CMA, UMD	0.190	0.208	0.205	0.236	0.228	0.159	0.152
Mkt, SMB, HML_S, RMW_{CS}, CMA_S, UMD_S	0.226	0.244	0.241	0.274	0.265	0.194	0.186
Mkt, S–F, H–F, R_C–F, C–F, U–F	0.177	0.193	0.191	0.218	0.212	0.147	0.141
Mkt, S–F, H_S–F, R_{CS}–F, C_S–F, U_S–F	0.188	0.204	0.201	0.229	0.221	0.159	0.153
Mkt, S–F,L–F, W_C–F, A–F,D–F	0.160	0.176	0.173	0.204	0.196	0.128	0.121
Mkt, S–F, L_S–F, W_{CS}–F, A_S–F, D_S–F	0.210	0.229	0.226	0.259	0.250	0.177	0.170

Panel B: Differences between $Sh^2(f)$ for cash and operating profitability factors

		Full-sample			In-sample			Out-of-sample		
Model	Actual	Average	Median	% < 0	Average	Median	% < 0	Average	Median	% < 0
Cash – Operating Profitability										
Mkt, SMB, HML, RMW, CMA, UMD	0.055	0.056	0.054	0.0	0.059	0.055	1.1	0.050	0.047	2.2
Mkt, SMB, HML_S, RMW_S, CMA_S, UMD_S	0.027	0.028	0.027	1.0	0.030	0.027	8.2	0.025	0.023	10.0
Mkt, S–F, H–F, R–F, C–F, U–F	0.043	0.045	0.043	0.0	0.047	0.044	1.1	0.041	0.039	2.0
Mkt, S–F, H_S–F, R_S–F, C_S–F, U_S–F	0.021	0.022	0.021	4.5	0.023	0.020	17.3	0.022	0.020	15.4
Mkt, S–F,L–F, W–F, A–F,D–F	0.048	0.049	0.047	0.2	0.052	0.047	4.1	0.043	0.040	5.0
Mkt, S–F, L_S–F, W_S–F, A_S–F, D_S–F	0.036	0.037	0.036	0.0	0.039	0.035	1.8	0.033	0.031	5.4

Results for Problem 2 and 3

- Table 4 is Distributions of differences between Sh2(f),

$$Sh^2(f_{column}) - Sh^2(f_{Row})$$

- Upward bias in FS and IS largely cancels in differences.

	Mkt, SMB, HML, RMW_C, CMA, UMD			*Mkt, SMB, HML_S, RMW_{CS}, CMA_S, UMD_S*			*Mkt, S–F, L_S–F, W_{CS}–F, A_S–F, D_S–F*		
Model	Average	Median	% < 0	Average	Median	% < 0	Average	Median	% < 0
Full-sample									
Mkt, SMB, HML, RMW_C, CMA, UMD				0.037	0.036	8.7	0.021	0.021	21.0
Mkt, SMB, HML_S, RMW_{CS}, CMA_S, UMD_S	−0.037	−0.036	91.3				−0.015	−0.015	82.5
Mkt, S–F, L_S–F, W_{CS}–F, A_S–F, D_S–F	−0.021	−0.021	79.0	0.015	0.015	17.5			
Mkt, SMB, HML, RMW_O, CMA, UMD	0.056	0.054	0.0	0.093	0.091	0.0	0.077	0.076	0.0
Mkt, S–F, H–F, R_C–F, C–F, U–F	0.015	0.014	24.6	0.051	0.050	3.1	0.036	0.035	12.3
Mkt, S–F, H_S–F, R_{CS}–F, C_S–F, U_S–F	0.004	0.003	45.7	0.040	0.040	2.8	0.025	0.025	21.4
Mkt, S–F,L–F, W_C–F, A–F,D–F	0.032	0.031	0.6	0.068	0.068	1.4	0.053	0.052	3.1
In-sample									
Mkt, SMB, HML, RMW_C, CMA, UMD				0.038	0.036	21.5	0.023	0.022	31.8
Mkt, SMB, HML_S, RMW_{CS}, CMA_S, UMD_S	−0.038	−0.036	78.5				−0.015	−0.014	69.6
Mkt, S–F, L_S–F, W_{CS}–F, A_S–F, D_S–F	−0.023	−0.022	68.2	0.015	0.014	30.4			
Mkt, SMB, HML, RMW_O, CMA, UMD	0.059	0.055	1.1	0.097	0.092	0.9	0.082	0.078	1.7
Mkt, S–F, H–F, R_C–F, C–F, U–F	0.018	0.016	31.5	0.056	0.051	13.7	0.041	0.038	23.7
Mkt, S–F, H_S–F, R_{CS}–F, C_S–F, U_S–F	0.007	0.006	45.5	0.045	0.043	11.8	0.030	0.029	29.7
Mkt, S–F,L–F, W_C–F, A–F,D–F	0.032	0.031	7.4	0.070	0.068	10.1	0.055	0.054	13.8
Out-of -sample									
Mkt, SMB, HML, RMW_C, CMA, UMD				0.035	0.033	20.0	0.018	0.017	33.5
Mkt, SMB, HML_S, RMW_{CS}, CMA_S, UMD_S	−0.035	−0.033	80.0				−0.017	−0.016	74.1
Mkt, S–F, L_S–F, W_{CS}–F, A_S–F, D_S–F	−0.018	−0.017	66.5	0.017	0.016	25.9			
Mkt, SMB, HML, RMW_O, CMA, UMD	0.050	0.047	2.2	0.085	0.081	1.1	0.068	0.064	3.1
Mkt, S–F, H–F, R_C–F, C–F, U–F	0.012	0.010	35.5	0.047	0.043	14.5	0.030	0.027	27.7
Mkt, S–F, H_S–F, R_{CS}–F, C_S–F, U_S–F	−0.001	−0.001	51.0	0.034	0.033	15.4	0.017	0.016	36.5
Mkt, S–F,L–F, W_C–F, A–F,D–F	0.031	0.029	7.6	0.066	0.062	7.8	0.049	0.046	12.5

Deconstructing $Sh^2(f)$

For each factor i, regress it on the other five model factors:

$$i_t = a_i + b_i^\top f_{-i,t} + e_{i,t}.$$

$$a_i = \text{spanning intercept}, \qquad \sigma_i^2 = (e_{i,t}).$$

The marginal contribution of factor i is:

$$\boxed{\frac{a_i^2}{\sigma_i^2} = \mathrm{Sh}^2(f) - \mathrm{Sh}^2(f_{-i})}$$

- a_i: average return of factor i unexplained by the other factors.
- σ_i^2: residual risk of factor i after controlling for the other factors.
- $\frac{a_i^2}{\sigma_i^2}$: conditional increase in $\mathrm{Sh}^2(f)$ from adding factor i.

A factor matters when it has a large unexplained mean return and low residual risk.

Deconstructing $Sh^2(f)$

Table 5
Spanning regressions and marginal contributions to $Sh^2(f)$ for three models that produce highest $Sh^2(f)$ in Table 4: July 1963–June 2016.

Mkt is *M–F*, the monthly excess return on the value-weight market portfolio. The size spread factor, *SMB*, is the difference between small stock and big stock portfolio returns (*S* and *B*). The value, investment, momentum, and cash profitability spread factors (*HML*, *CMA*, *UMD*, and RMW_C) are averages of small stock and big stock spread factors (HML_S, CMA_S, UMD_S, RMW_{CS} and HML_B, CMA_B, UMD_B, RMW_{CB}). L_S–*F*, W_{CS}–*F*, A_S–*F*, and D_S–*F* are excess returns on the short ends of the small stock spread factors, HML_S, CMA_S, RMW_{CS}, and UMD_S. For the three models that produce the highest $Sh^2(f)$ in Table 4, the table shows intercepts *a*, *t*-statistics for the intercepts $t(a)$, slopes, R^2, and residual standard errors $s(e)$ from spanning regressions of each of the factors of a model on the model's other five factors. The table also shows $Sh^2(f)$ and each factor's marginal contribution to a model's $Sh^2(f)$, that is, $a^2/s^2(e)$.

Panel A: Spanning regressions for combined spread factor model

LHS	*a*	*Mkt*	*SMB*	*HML*	RMW_C	*CMA*	*UMD*	$t(a)$	R^2	$s(e)$	$Sh^2(f)$	$a^2/s^2(e)$
Mkt	1.04		0.06	0.04	−0.87	−0.70	−0.07	6.77	0.31	3.70	0.190	0.080
SMB	0.48	0.03		0.04	−0.83	0.03	0.06	4.35	0.27	2.59	0.190	0.034
HML	0.08	0.01	0.03		0.16	0.94	−0.13	0.90	0.51	1.95	0.190	0.002
RMW_C	0.43	−0.14	−0.27	0.09		−0.02	0.06	7.00	0.41	1.47	0.190	0.085
CMA	0.17	−0.10	0.01	0.47	−0.02		0.04	2.87	0.53	1.38	0.190	0.015
UMD	0.61	−0.09	0.14	−0.53	0.46	0.34		3.55	0.10	4.02	0.190	0.023

Panel B: Spanning regressions for small stock spread factor model

LHS	*a*	*Mkt*	*SMB*	HML_S	RMW_{CS}	CMA_S	UMD_S	$t(a)$	R^2	$s(e)$	$Sh^2(f)$	$a^2/s^2(e)$
Mkt	1.02		0.18	−0.16	−0.36	−0.51	−0.13	6.19	0.23	3.88	0.226	0.070
SMB	0.45	0.08		0.03	−0.60	0.09	−0.01	4.03	0.27	2.60	0.226	0.031
HML_S	−0.05	−0.04	0.01		0.69	0.84	−0.09	−0.59	0.64	1.91	0.226	0.001
RMW_{CS}	0.42	−0.07	−0.24	0.51		−0.42	0.04	6.03	0.56	1.64	0.226	0.066
CMA_S	0.30	−0.07	0.03	0.48	−0.32		0.05	4.73	0.47	1.45	0.226	0.042
UMD_S	0.96	−0.14	−0.03	−0.42	0.22	0.37		5.41	0.05	4.12	0.226	0.054

Panel C: Spanning regressions for small stock short factor model

LHS	*a*	*Mkt*	*S–F*	L_S–*F*	W_{CS}–*F*	A_S–*F*	D_S–*F*	$t(a)$	R^2	$s(e)$	$Sh^2(f)$	$a^2/s^2(e)$
Mkt	−0.03		0.71	0.19	−0.66	0.42	0.03	−0.41	0.81	1.93	0.210	0.000
S–F	0.31	0.10		−0.27	0.53	0.52	0.04	10.50	0.98	0.73	0.210	0.179
L_S–*F*	0.07	0.04	−0.44		0.55	0.88	−0.05	1.81	0.98	0.93	0.210	0.006
W_{CS}–*F*	−0.19	−0.13	0.71	0.45		−0.02	0.02	−5.11	0.98	0.85	0.210	0.048
A_S–*F*	−0.12	0.06	0.48	0.50	−0.01		0.04	−4.06	0.99	0.70	0.210	0.031
D_S–*F*	−0.52	0.07	0.65	−0.41	0.21	0.60		−4.30	0.85	2.79	0.210	0.034

Figure: Spanning regressions and marginal contributions to $Sh^2(f)$

Performance Measures

Each row is a six-factor model. A denotes the average across all LHS test portfolios.

Column	Interpretation
GRS, $p(\mathrm{GRS})$	Joint test of whether all pricing errors are zero. Lower GRS and higher p-value are better.
$A\lvert a_i\rvert$	Average absolute alpha. Lower is better.
$\frac{Aa_i^2}{A\bar{r}_i^2}$	Average squared alpha, scaled by average squared portfolio return. Lower is better.
AR^2	Average regression R^2 across LHS portfolios. Higher is better.
$\mathrm{Sh}^2(f)$	Maximum squared Sharpe ratio of model factors. Higher is better; independent of LHS portfolios.
$\mathrm{Sh}^2(a)$	Maximum squared Sharpe ratio of unexplained alphas. Lower is better.

Comparison with other measures

Table 7
Summary statistics for regression intercepts, July 1963–June 2016.

The 125 left-hand-side (LHS) portfolios in Panel B are from five 5×5 quintile sorts on *ME* and, independently, on *BE/ME*, *OP*, *CP*, *Inv*, or *Mom*. (See Appendix for definitions of all variables.) Like the 2×3 portfolios used to construct the factors, the *ME–Mom* portfolios are formed monthly and the other portfolios in Panel B are formed at the end of June. Eighty-five of the anomaly portfolios in Panel C are from independent, end-of-June, 5×5 sorts on *ME* and market β, 5×5 sorts on *ME* and accruals, and 5×7 sorts on *ME* and net issuance (repurchases, zero net share issues, and quintiles of positive net share issues). The remaining 50 anomaly portfolios are from monthly 5×5 sorts on *ME* and either *Var*, the variance of daily returns, or *RVar*, the variance of daily residuals from the Fama and French (1993) three-factor model. The breakpoints for *Var* and *RVar* quintiles are conditional on *ME* quintile. The 260 LHS portfolios in Panel A are all the LHS portfolios in Panels B (125) and C (135). The table shows the *GRS* statistic of Gibbons et al. (1989) and its *p*-value, $p(GRS)$; $A|\alpha_i|$, the average absolute intercept; $A\alpha_i^2/A\bar{r}_i^2$, the average squared intercept over the average squared value of $\bar{r}_i$, which is the difference between the average return on LHS portfolio i and the average return on the value-weight market; $As^2(\alpha_i)/As_i^2$, the average of the squared sample standard errors of the intercepts over the average squared intercept; the average of the regression R^2, AR^2; $Sh^2(f)$, the maximum squared Sharpe ratio for the model's factors; and $Sh^2(a)$, the max squared Sharpe ratio for the intercepts for a set of LHS portfolios.

Model	*GRS*	$p(GRS)$	$A\|a\|$	$A\alpha_i^2/A\bar{r}_i^2$	$As^2(a_i)/As_i^2$	AR^2	$Sh^2(f)$	$Sh^2(a)$
Panel A: All portfolios in Panels B and C								
Mkt, SMB, HML$_5$, *RMW*$_{C5}$, *CMA*$_5$, *UMD*$_5$	2.26	0.000	0.087	0.14	0.42	0.91	0.226	1.926
Mkt, S–F, L$_2$*–F, W*$_{C5}$*–F, A*$_5$*–F, D*$_5$*–F*	2.33	0.000	0.103	0.18	0.32	0.91	0.210	1.958
Mkt, SMB, HML, RMW$_C$*, CMA, UMD*	2.35	0.000	0.095	0.18	0.33	0.91	0.190	1.943
Mkt, S–F, H$_5$*–F, R*$_{C5}$*–F, C*$_5$*–F, U*$_5$*–F*	2.39	0.000	0.111	0.28	0.22	0.91	0.188	1.975
Mkt, S–F, H–F, R$_C$*–F, C–F, U–F*	2.40	0.000	0.106	0.25	0.25	0.90	0.177	1.961
Mkt, S–FL–F, W$_C$*–F, A–FD–F*	2.47	0.000	0.094	0.19	0.31	0.91	0.160	1.989
Mkt, SMB, HML, RMW$_D$*, CMA, UMD*	2.54	0.000	0.094	0.19	0.30	0.91	0.135	2.010
Panel B: 5 × 5 sorts on ME and BE/ME, OP, CP, Inv, and Mom								
Mkt, SMB, HML$_5$, *RMW*$_{C5}$, *CMA*$_5$, *UMD*$_5$	1.99	0.000	0.072	0.09	0.58	0.92	0.226	0.599
Mkt, S–F, L$_2$*–F, W*$_{C5}$*–F, A*$_5$*–F, D*$_5$*–F*	2.07	0.000	0.091	0.14	0.35	0.92	0.210	0.613
Mkt, SMB, HML, RMW$_C$*, CMA, UMD*	2.17	0.000	0.082	0.13	0.38	0.92	0.190	0.631
Mkt, S–F, H$_5$*–F, R*$_{C5}$*–F, C*$_5$*–F, U*$_5$*–F*	2.22	0.000	0.084	0.14	0.37	0.92	0.188	0.647
Mkt, S–F, H–F, R$_C$*–F, C–F, U–F*	2.24	0.000	0.081	0.14	0.36	0.92	0.177	0.645
Mkt, S–FL–F, W$_C$*–F, A–FD–F*	2.31	0.000	0.086	0.15	0.33	0.92	0.160	0.657
Mkt, SMB, HML, RMW$_D$*, CMA, UMD*	2.43	0.000	0.079	0.12	0.37	0.92	0.135	0.675
Panel C: 5 × 5 sorts on ME and accruals, β, variance of daily returns and residuals, and 5 × 7 sorts on ME and net share issuance								
Mkt, SMB, HML$_5$, *RMW*$_{C5}$, *CMA*$_5$, *UMD*$_5$	2.62	0.000	0.100	0.19	0.36	0.90	0.226	0.867
Mkt, S–F, L$_2$*–F, W*$_{C5}$*–F, A*$_5$*–F, D*$_5$*–F*	2.63	0.000	0.114	0.22	0.30	0.90	0.210	0.858
Mkt, SMB, HML, RMW$_C$*, CMA, UMD*	2.66	0.000	0.106	0.23	0.31	0.89	0.190	0.853
Mkt, S–F, H$_5$*–F, R*$_{C5}$*–F, C*$_5$*–F, U*$_5$*–F*	2.80	0.000	0.136	0.41	0.17	0.89	0.188	0.899
Mkt, S–F, H–F, R$_C$*–F, C–F, U–F*	2.82	0.000	0.130	0.35	0.20	0.89	0.177	0.895
Mkt, S–F, L–F, W$_C$*–F, A–F, D-F*	2.71	0.000	0.102	0.23	0.30	0.89	0.160	0.849
Mkt, SMB, HML, RMW$_D$*, CMA, UMD*	2.88	0.000	0.108	0.25	0.26	0.90	0.135	0.881

Figure: Summary statistics for regression intercepts

Comparison with other measures

- $\mathrm{Sh}^2(f)$ is invariant to the choice of test portfolios, because it depends only on model factors.
- GRS, $\mathrm{Sh}^2(a)$, and $\mathrm{Sh}^2(f)$ tend to give similar rankings, but this agreement is partly mechanical: they are all based on the same GRS/Sharpe-ratio framework.
- The model that wins on $\mathrm{Sh}^2(f)$ also performs best on the equal-weighted alpha measures considered in the paper.
- Beyond the top model, rankings differ across performance measures: a model that is strong on $\mathrm{Sh}^2(f)$ need not be strong on equal-weighted pricing errors.

$\mathrm{Sh}^2(f)$ is a useful ranking criterion, but not the unique definition of a better model.

References I

Barillas, F. and Shanken, J. (2017). Which alpha? *The Review of Financial Studies*, 30(4):1316–1338.

Gibbons, M. R., Ross, S. A., and Shanken, J. (1989). A test of the efficiency of a given portfolio. *Econometrica*, 57(5):1121–1152.

Thank You